

15-4-2025

F.D (revision):

Student → ERD → Schema

Analyze Design → tables

Informal Guidelines

i) No insertion, updation, deletion, anomalies

ii) No Nulls.

Formal Guidelines

F.D

Partial

Fully

Composite

Determinant

XY → Z

Only one uniquely identifies

Both are identifying

Z

Normalization

"Reduce bad relations/tables in multiple tables/relation"

1st Normal Form (1NF):

↳ Multi Valued Attribute

↳ Composite Attribute

↳ Nested Relation

No atomic Value

Example:

Department (D-ID, Dname, Dmgr, Dlocation)

(MVA)

1	Admin	145	KAI, LAB, ISB
2	Resuh	125	PWR, RWP, LAR

↳ Sql allows normalization thus these errors are not in sql server. but in python dictionaries database.

Normalization:

→ "Good Design"

→ Bad Relations

→ 1st NF ✓

→ 2nd NF ✓

→ 3rd NF ✓

→ 4th NF } not in course

→ 5th NF }

→ BCNF ✓

↳ 1NF excluded if (no atomic value) exists:

2	Resuh	125	PWR
2	Resuh	125	LAR

0) Separate Rows → redundancy of data, not considered

0) Separate table → Department No/locations as Primary Key's

0) Separate Columns. → lot of nulls.

↳ These are same tests for tables.

Second Normal Form (2NF):

- ① Table is already in 1NF
 ② "The relation should not have any Partial Dependency"

↳ to look composite keys.
 (Composite determinant.)

Redundant tables of 2NF

SSN | PNO | Hous

SSN | EName

PNO | PName | Plocatn

Attr1	Attr2	SSN Ename	Kloc

⌘

Third Normal Form: (3NF)

① Table is in 2NF

② no Non-prime attribute 'A' in R is transitively dependent on P.K.

$X \rightarrow Y$; $Y = \text{non-prime}$.

$Y \rightarrow Z$

$X \rightarrow Z$

If both are uniquely identifying then in 2NF else not.

Already in 1NF

But has insertion, deletion, update anomaly

SSN | PName | Hous | Ename | PName | Plocatn

1st step: Write all possible F.D's

i) SSN, PName $\rightarrow$ Hous $\checkmark$ (Fully)

ii) $\checkmark$, " $\rightarrow$ Ename $\times$ (Partial only SSN)

iii) " , " $\rightarrow$ PName $\times$ (Partial only PName)

iv) $\checkmark$, " $\rightarrow$ Plocatn $\times$ (Partial only PName)

Fully F.D's

SSN, PName $\rightarrow$ Hous

SSN $\rightarrow$ Ename

PName $\rightarrow$ PName

PName $\rightarrow$ Plocatn

Then make tables on basis of these fully F.D's

Ename SSN Plocatn Address Dnum Dname Dnum

↳ Here used arrow FD representation.

SSN $\rightarrow$ Dnum

Dnum $\rightarrow$ Dname

SSN $\rightarrow$ Dname

Non-Prime is indirectly involved.

Not in 3NF

→ Pnum is a non-key
but acting as a
Key (uniquely identifies)

Make separate table
for it.

Example:

but supplier supplies in only 1 location

Sup#	Part#	lotID	locName	Qty
S1	1	10	London	1600
S1	2	10	London	200
S2	2	20	Paris	500
S3	3	30	NewYork	3000

Sup# | Part# | Qty | Supply-2NF

Sup# | loc# | locName | Supply-2NF

3NF

Sup# → loc#

loc# → locName

~~locName → Sup#~~

Sup# → locName

Sup# | Part# | Qty | Supply-3NF

Sup# | locID | Supply-loc 3NF

locID | locName | loc-3NF

Anomalies:

→ Insert: (1 supplier wants to supply
but has not yet supplied)

→ P-Key: (1 supplier supplies same
(Update) part to two locations
cant be prevent)

→ Delete: what if we want
to delete part.

Example:

SSKSSN StdClass CF

1NF ✓

2NF ✓ Sup → locID ✓

Sup → locName ✓

Sup, Part → Qty ✓

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Subj: Prof

stdid: ~~Prof~~

BCNF

↳ "Non-prime attribute can't identify a unique/prime-attribute".

stdid	subject	Prof. id
2	Java	p.java Ali
4	C++	p.cpp Sara
1	J++	p.java2 Aleem
3	C++	p.c++ Sidra
1	C++	p.cpp2 Ali

↳ check partially dependencies

stdid, subject $\rightarrow$ Prof. id

↳ no partial $\rightarrow$ fully F.D

it is in 1st 2nd 3rd NF

In this case

Prof. id identifies

course such as Java course which is instructed by Ali or Aleem.

↳ it's not in BCNF.

↳ Now, what to do!!!

↳ separate table.

Example: Normalization

	Prop id	Candidate Key County Lot #		Area	Price	Tax rate.
lot A	↑	↑	↑	↑	↑	↑

} County, lot → base
 (Part)
 } only tax depends
 county. (so 2NF X)

Constraints:

- ① Area-price fixed: 10 marla → 5 lakh
- ② County tax rate fixed (Tehsil / District = County) ^{sector}
- ③ County A → 15, 16, 17
- ④ County B → 18, 19, 20

note candidate key is not a non-prime, it is prime attribute.

lot 2NF

lot #	County	tax rate
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3NF Area → Price (nonprime → nonprime)

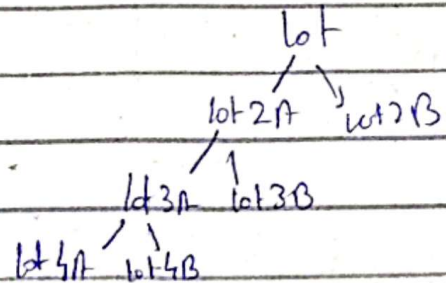
So, we remove the acting like prime attribute.

⇒ lot 3A

Prop id	County	lot #	Area
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⇒ lot 3B

Area	Price.
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Now, BCNF Area → County (As in constraints Area is fixed for two diff. counties which identifies county by size)

⇒ lot 4A

Prop id	lot #	Area
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⇒ lot 5A

Area	County
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note: we remove the one identifying.